Improving Motor Imagery EEG Signal Classification Using Deep Learning Techniques

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***Abstract*——MotorImagery-based Brain-Computer Interfaces (MI-BCIs) facilitate direct neural communication with external systems by decoding imagined movements from electroencephalogram (EEG) signals. Despite their transformative potential in neuroprosthetics, rehabilitation, and assistive technologies, the classification of MI-EEG signals remains a formidable challenge due to their low signal-to-noise ratio, inter-subject variability, and the presence of physiological artifacts. This study proposes an innovative deep learning architecture that harmoniously integrates a Variational Autoencoder (VAE), a Deep Autoencoder (DAE), and a hybrid Convolutional Neural Network-Long Short Term Memory (CNN-LSTM) model to address these limitations. The VAE serves as a robust denoising mechanism, attenuat ing noise while preserving the latent representations of neural dynamics. Subsequently, the DAE extracts high-level, abstract features that encapsulate both spatial and frequency-domain information vital for discriminating between MI tasks. These features are then processed by a CNN-LSTM network that captures spatial hierarchies via convolutional filters and models temporal dependencies through LSTM units. This synergistic approach enables the network to learn complex spatiotemporal patterns inherent in MI-EEG signals. The proposed model was rigorously evaluated on a multi-class MI dataset, achieving a superior classification accuracy of 94.36%, along with notable improvements in precision, recall, and F1-score compared to existing benchmarks. The results substantiate the efficacy of the hybrid architecture in enhancing MI classification and under line its potential for deployment in real-time BCI applications. This work contributes a robust, generalized and noise-resilient framework.**

***Index Terms*—Variational Autoencoder (VAE), Deep Autoencoder (DAE), Convolutional Neural Network (CNN), Electroencephalogram (EEG), Brain-computer Interface (BCI), Motor Imagery (MI).**

I. INTRODUCTION

Electroencephalography (EEG), a noninvasive and inexpensive technology to record brain activity, is the principal channel for Motor Imagery (MI) MI-based BCI systems owing to its high temporal resolution and easy usage. However, EEG readings are naturally noisy due to physical and environmental disturbances, making correct classification of MI tasks difficult. Traditional machine learning methods, such as Common Spatial Patterns (CSP) and Support Vector Machines (SVM), have been frequently used in MI categorization. Additionally, time-frequency analysis approaches such as Short-Time Fourier Transform (STFT) and Wavelet Transform have been investigated for feature extraction. Despite these developments, standard approaches struggle to extract features from noisy and high-dimensional EEG inputs.

In this model Motor Imagery EEG data was collected to ensure more control over subject variability, data quality, and experimental setup. The data set collected is limited to motor imagery tasks using the left and right hands, as well as the left and right feet that presents a deep learning-based pipeline which combines VAE for noise reduction, DAE for feature extraction, and a hybrid CNN-LSTM model for the classification of EEG signals from motor imagery, resulting in around 94.36% accuracy.

II. LITERATURE SURVEY

Recent research proved the efficacy of deep learning models for MI classification. A multibranch convolutional neural network containing attention mechanisms (MBEEGCBAM) enhanced classification precision by highlighting important spatial and temporal characteristics of EEG data, obtaining 87.2% [1]. In a different study, an original approach incorporated a local reparameterization trick (LRT) into the convolutional neural network (CNN) architecture which enhanced the generalization performance of the model with an accuracy of 99.79% [2]. Many researchers have focused on hybrid models that integrate spatial and temporal feature extraction. One recent study used a deep recurrent convolutional neural network (RCNN) to convert MI EEG data to 2D images which were later processed by a CNN-LSTM network achieving a 70.64% classification accuracy [3]. To overcome the issue of limited EEG datasets, data augmentation techniques have been frequently used. One research used empirical mode decomposition (EMD) to create synthetic EEG samples, which greatly improved the performance of the model in short data set settings [4]. The authors in a paper [5] proposed a predictive model framework that can get consumer choice towards E-commerce products in terms of "likes" and "dislikes" by analyzing EEG signals.

Beyond categorization, researchers have concentrated on real-world BCI applications. One research created an anatomy-related robotic hand driven by MI-EEG data, using a basic eight-electrode system for signal collection [6]. This study established the practical use of MI classification for prosthesis control, with an accuracy of 86.2%. In [7]to perform preprocessing, the authors used common average reference (CAR) filtering and Laplace filtering of EEG signals. Some works [8] [9] [10] have developed their models based on deep learning to perform motor imagery classification. The literature review emphasizes the ongoing development of deep learning approaches in MI-EEG categorization. Various procedures for improving classification accuracy have been investigated by incorporating insights from current studies [11], such as hybrid CNN-LSTM architectures, enhanced methods for obtaining features, and data augmentation approaches. These methods address issues such as poor signal-to-noise ratio (SNR), subject-specific variability, and the scarcity of EEG datasets. some recent works [12], [13], [14] have explored multi-task learning (MTL) in MI-EEG classification. The proposed framework, which combines VAE for noise reduction, DAE for feature extraction, and CNN-LSTM for classification, builds on these advances to improve MI classification accuracy even further. The model intends to increase robustness and generalizability across various disciplines by adding noise-filtering methods and deep feature extraction.

III. Methodology

A. Data Collection

This dataset focuses solely on four MI tasks: left-hand, right-hand, left-foot, and right-foot imaginary tasks, reducing superfluous noise. Data dependability is increased by regulating the gathering setting and ensuring that participants completely understood the activities.

As shown in Figure 1 and Figure 2 typical data collection was made at the Chinta Neuro Center which is located in Hanamkonda, Telangana, India. The data was collected with the informed consent of the participants. Around 16 participants gave the MI EEG recordings. These EEG signals were collected using the PsychoPy software, which allowed for controlled stimulus presentation and response synchronization, resulting in accurate and reliable data gathering. EEG data was obtained utilizing a multi-channel headset that followed the 10-20 international electrode placement method to ensure complete scalp coverage. Furthermore, data was collected from a varied range of subjects, which improved generalizability.

Fig. 1. Data acqusition

B. Dataset Overview

The dataset includes 160 occurrences and 27 characteristics that characterize different qualities of EEG signals, primarily for Motor Imagery (MI) classification. These characteristics are produced using a variety of techniques, including statistical, spectral, and wavelet-based approaches. The signal is analysed using Hjorth parameters, which is considered as one of the vital features. Mobility monitors the frequency dynamics of the signal, whereas Complexity evaluates the unpredictability or disorder in the signal's frequency content. Kurtosis and skewness can provide statistical insights into the spread of EEG data, highlighting the form and asymmetry of the signal distribution. The dataset also contains First and Second Difference features. The initial difference characteristics, such as Mean and Max, assess the pace at which the EEG signal varies. Finally, the Epoch function identifies certain segments or sections of the EEG data, allowing for more precise study of the signal across time. These different properties give a rich and complete representation of the EEG signal, making the dataset appropriate for tasks such as MI classification and other EEG-based analysis.

C. Data Preprocessing

It is important to preprocess the data that has been collected and to preprocess the data following steps are done.

- Signal Acquisition and Artifact Reduction As EEG data is susceptible to interference from muscle activity, eye blinks, and external electrical interference, several preprocessing approaches were used to address these concerns. A band-pass filter (0.5-45 Hz) was used to maintain key frequency components associated with MI activities,

Fig. 2. Data acquisition

notably mu (8-12 Hz) and beta (13-30 Hz) oscillations, which are suggestive of motor-related brain activity [6].

- Channel Selection and Standardization To increase classification accuracy, EEG channels positioned over the motor cortex were chosen since previous research has identified them as critical areas for MI-based brain-computer interfaces (BCIs).
- Feature Extraction Methods Important techniques such as elastic net and Common Spatial Pattern (CSP) have shown the effectiveness in identifying most relevant EEG features and improving classification accuracy [11], [15].
- Dimensionality Reduction and Signal Representation. To deal with the high-dimensional character of EEG data, Principal Component Analysis (PCA) was used to reduce duplicate features while maintaining important information [16].

D. FLOW CHART

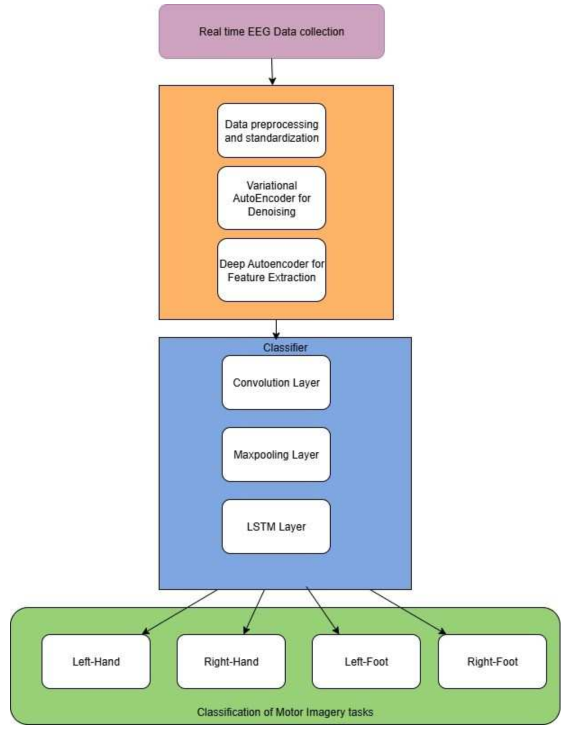

Fig. 3. Flow chart representing work flow

Fig. 3 indicates the flowchart of how the classification of motor imagery EEG is done from data collection to classification. Where first the EEG data containing information about motor imagery tasks such as left hand movement, right hand movement, left-foot movement, and right-foot movement imagination is collected from the subjects. The data is then pre-processed where in the dimensionality reduction, feature extraction, noise removal and data augmentation is done.The pre-processed data is then given to variational autoencoder where the noise such as noise created by muscle movements and noise due to improper placement of electrodes is removed.Then this denoised data is sent to a deep autoencoder where the deep autoencoder using its encoder and decoder extracts important features that may be useful for classification and discard irrelevant features. Then the compressed features are taken as input for the convolutional neural networks-long short-term memory hybrid model where temporal and spatial dependencies are extracted patterns are identified and the data is classified into the motor imagery classes such as left-hand, right-hand,left-foot and right foot based on input.

IV. Model Training

A. Variational Autoencoder (VAE)

Variational Autoencoder (VAE) efficiently learns compact representations of input data while maintaining important patterns. VAEs are important in EEG-based Brain-Computer

Interfaces (BCIs) because they reduce noise from movements of muscles, eye blinks, and other anomalies while preserving critical motor-related data required for categorization. Fig. 4

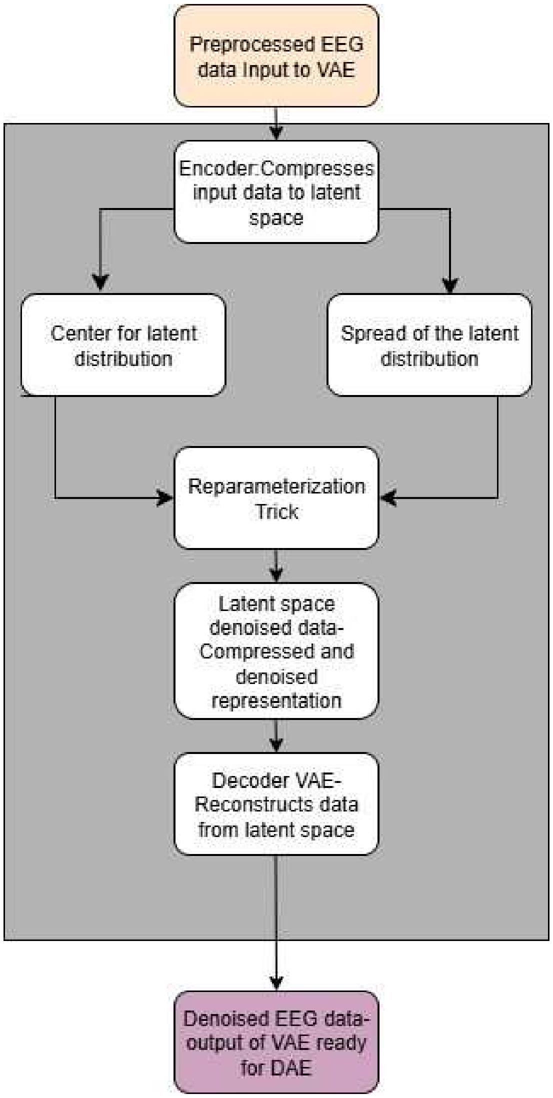

Fig. 4. Flowchart showing denoised EEG data

represents flowchart of VAE where its design includes an encoder, a latent space transformation, and a decoder, which enable it to learn an efficient representation of EEG signals. The encoder takes pre-processed EEG data and returns two statistical parameters: the mean, which defines the centre of distribution, and the log variance, which shows the distribution's spread. In place of using a fixed value, VAE employs a probabilistic sampling strategy in which a latent vector is produced from the mean and variance, guaranteeing that the learnt representations are smooth and generalizable. The decoder reconstructs the EEG signal using the latent space representation, to provide a denoised version of what was originally input. This method guarantees that important motor imagery information is maintained while undesirable noise is successfully reduced. By this the model guarantees that the motor-related aspects of EEG data are preserved while reducing noise.

B. Deep Autoencoder (VAE)

[12] proposed a hybrid model by combining a Variational Autoencoder (VAE) for noise reduction with a Deep Autoencoder (DAE) and Convolutional Neural Network (CNN) for classification. Deep autoencoder (DAE) compresses and reconstructs input data efficiently. It is used to fine-tune the feature representations generated by the Variational Autoencoder (VAE), making sure the most significant properties are kept while minimizing dimension. Fig. 5 represents the

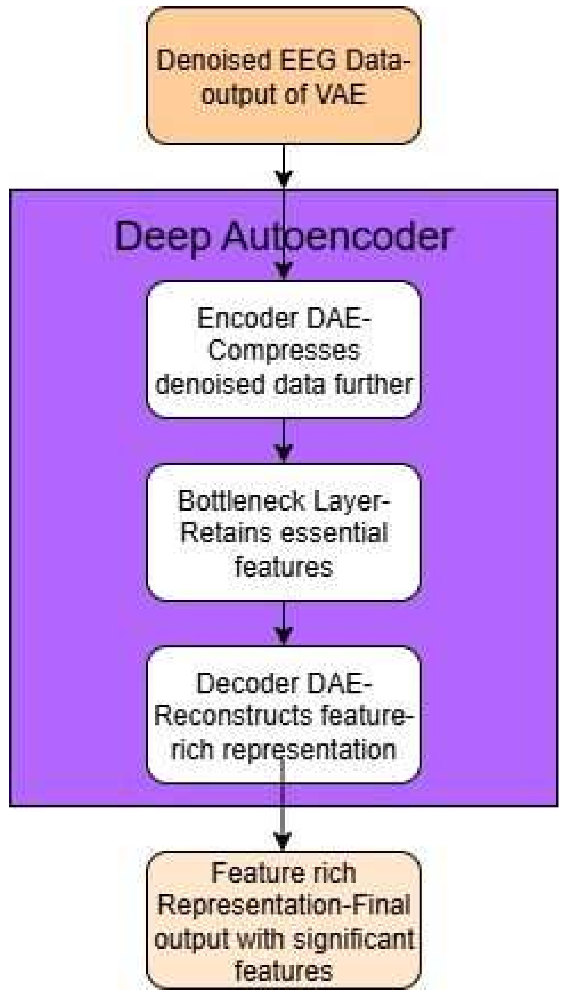

Fig. 5. Flowchart representing feature extraction

flowchart of DAE. This allows the model to reject redundant information while retaining the most important characteristics for classification. Once trained, the DAE's encoder extracts compressed feature representations these characteristics provide highly meaningful inputs for subsequent classification tasks, increasing the reliability of motor imagery-based EEG classification.

C. Hybrid Convolutional Neural Network (CNN) with Long Short-Term Memory (LSTM)

The CNN+LSTM hybrid model classifies motor imagery (MI) EEG data using both spatial and temporal information retrieved from the Deep Autoencoder's compressed representations. This model successfully combines the feature extraction skills of Convolutional Neural Networks (CNNs) with the sequential pattern-learning capacity of Long Short-Term Memory (LSTM) networks, making it ideal for EEG-based Brain-Computer Interface (BCI) applications. The initial processing level involves CNN layers that extract spatial characteristics from DAE-encoded EEG representations. The convolutional layers use filters to find patterns of spatial associations in the data, increasing key signal components and decreasing noise. The collected spatial data are then flattened and sent into the LSTM network for temporal analysis. The LSTM layers analyse the sequential feature representations produced by CNNs, learning the temporal relationships between distinct time steps in the data. The final classification layer is made up of a fully connected (dense) layer followed by a Softmax activation function that generates the probability distribution over MI classes. The CNN+LSTM architecture accurately

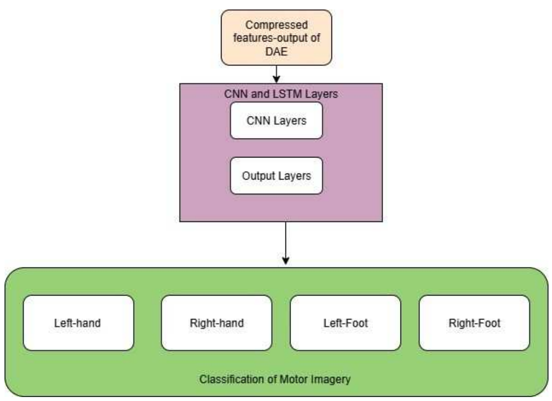

Fig. 6. Flowchart representing classification of Motor Imagery signals

identifies MI EEG data by integrating CNNs and LSTMs model. Fig. 6. represents Flowchart of CNN+LSTM hybrid model this hybrid model guarantees that spatial and temporal connections are learnt, hence improving the overall accuracy of MI classification.

D. Model Evaluation

The efficiency of the proposed model for motor imagery (MI) EEG classification is assessed using a train-test split technique, with the dataset separated into training and testing subsets. Z-score normalization was then used to normalize the data which helps to eliminate bias. Artifact removal approaches improved EEG data quality by removing noise. Model performance is assessed using the accuracy, confusion matrix, and classification report, which revealed how effectively the model distinguished between motor imagery tasks and its capacity to generalize on the test set. The fig. 7

Model	Accuracy	Precision	Recall	F1-Score
LDA	78.2%	0.79	0.76	0.775
SVM	81.6%	0.80	0.82	0.810
RF	78.5%	0.79	0.78	0.785
KNN	74.3%	0.72	0.74	0.730
CNN+DAE	85.7%	0.84	0.86	0.850
Proposed Method	94.36%	0.94	0.947	0.940

Table. I. comparison of models performance.

indicates comparison of Various models in MI EEG Classification which shows that the suggested model has an accuracy of 94.36%, beating standard techniques like LDA (78.2%), Random Forest (78.5%), and KNN (74.3%). The CNN model alone obtained 85.7% accuracy, but the addition of VAE and LSTM boosted feature extraction, boosting classification performance by collecting both spatial and temporal EEG signal features. The proposed model's high recall demonstrates its ability to properly categorize MI signals, lowering the possibility of misunderstanding in real-time BCI applications.

V. Results and Analysis

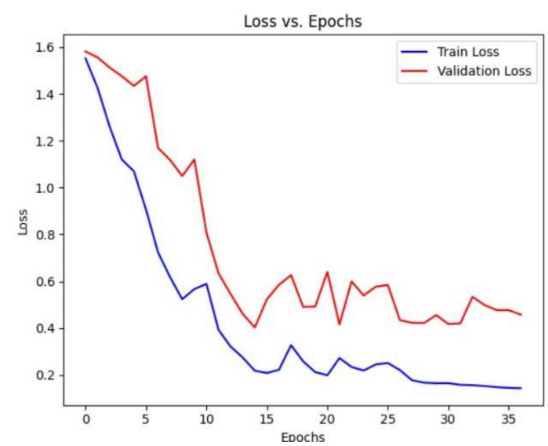

Fig. 7. Plot of loss vs epochs

An epoch represents a complete transit of the dataset through the model. Fig. 8 represents plot of loss vs epochs where increase in epochs is observed during decrease in loss which indicates model's predictions are coming closer to true values. Fig. 9. represents plot of accuracy vs epochs As epochs

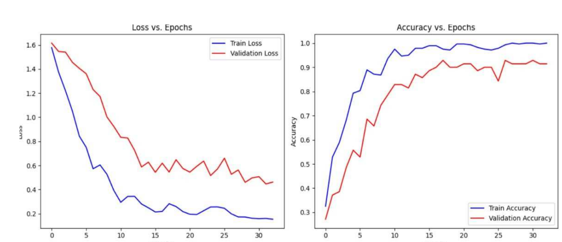

Fig. 8. Plot of accuracy vs epochs

increase accuracy also increases indicating better learning.

This demonstrates that the model is learning efficiently, optimizing feature extraction, and improving Motor Imagery EEG data classification.

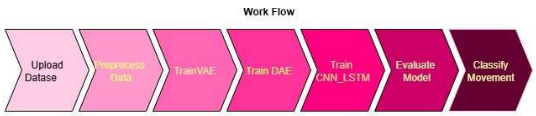

Fig. 9. represents Graphical User Interface

Fig. 10 represents Graphical User Interface it provides an easy framework for deep learning-based classification of motor imagery (MI) EEG signals. It allows users to upload and preprocess EEG data, removing noise and visually comparing raw and pre-processed signals. The system automates training for VAE, DAE, and CNN-LSTM models and displays accuracy and loss graphs to monitor performance.

VI. Discussion

The findings show that the proposed VAE + DAE + CNN + LSTM model improves motor imagery (MI) EEG signal categorization. The 94-95% accuracy gained much exceeds classic machine learning classifiers like LDA, KNN, SVM, and Random Forest, which normally reach 75-83% accuracy. The proposed model's exceptional performance stems from its ability to efficiently capture spatial, temporal, and non-linear EEG signal properties.

VII. Future Work

While the suggested model performs well in MI EEG . The computational complexity of deep learning models might hinder real-time BCI applications. Future research can focus on creating lightweight and streamlined architectures, such as edge AI implementations, to allow low-latency processing in real-world applications.Brain signals differ greatly between people. Future research can look at individualized deep learning models that adapt in real time utilizing reinforcement learning that guarantee that BCI systems are more user-centric and efficient.The suggested model may be evaluated and verified in real-time BCI applications such as brain-controlled robots and neurofeedback systems. Future research can focus on hardware integration to ensure that deep learning-based EEG categorization may be applied to wearable BCI devices for usage in real life.

VIII. Conclusion

This study proposes a deep learning system based on VAE + DAE + CNN + LSTM for the robust categorization of motor imagery (MI) EEG signals. The suggested technique improves feature extraction and classification performance by using a variational autoencoder (VAE) for initial feature learning and noise removal, deep autoencoders (DAE) for further compression of features, and a CNN-LSTM hybrid model to capture spatial-temporal relationships in EEG data and to classify motor imagery EEG signals. The findings show a considerable improvement in classification accuracy, reaching 94-95% .

In neurorehabilitation, the suggested model can help stroke patients regain motor control by understanding and converting brain impulses into movement orders. Furthermore, in the gaming and virtual reality (VR) industries, this method allows for thought-controlled interactions, resulting in a smooth and engaging user experience.

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